

Heilbronn's Triangle Problem: The Upper Bound

1 Definition

Let $U = [0, 1]^2$. For a finite set $P \subseteq U$, define

$$h(P) = \min_{\substack{x, y, z \in P \\ \text{distinct}}} \text{area}(\Delta(x, y, z)), \quad (1)$$

where $\Delta(x, y, z)$ denotes the triangle formed by the three points x, y, z , and degenerate triangles have area zero.

Definition 1 (The n th Heilbronn number). For $n \geq 3$, the n th *Heilbronn number* of the unit square is

$$H(n) = \max_{\substack{P \subseteq U \\ |P|=n}} h(P). \quad (2)$$

A set attaining this maximum is called an *optimal Heilbronn configuration*.

The maximum exists by compactness. The use of the unit square is convenient but not essential for asymptotic questions: for any two fixed convex bodies with nonempty interior, the corresponding Heilbronn numbers differ by multiplicative constants depending only on the two bodies.

2 Current best result

The best known asymptotic bounds are

$$H(n) = \Omega\left(\frac{\log n}{n^2}\right), \quad H(n) \leq n^{-7/6+o(1)}. \quad (3)$$

3 Problem formalization

Prove an asymptotic upper bound for $H(n)$ that is strictly stronger than

$$H(n) \leq n^{-7/6+o(1)}. \quad (4)$$